

Fourier Series and Partial Differential Equations

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Condensed Cheat Sheet: Fourier Coefficients a_n, b_n , Half-Range Series, Separation of Variables, the Laplacian $\nabla^2 u$, and the Heat/Wave Equations

Unit 18 at a Glance: Fourier Series as the Key to Partial Differential Equations

A **Fourier series** expresses a periodic function as a sum of sines and cosines, and because the trigonometric system is *orthogonal*, every coefficient is one independent integral — the **Euler–Fourier formulas**. Symmetry simplifies the work: **even** functions need only cosines, **odd** functions only sines, and a function on a half-interval can be expanded as either via a half-range extension. This is the toolkit that cracks **partial differential equations**: **separation of variables** $u = X(x)T(t)$ splits a PDE into ODEs linked by a separation constant, the boundary conditions reproduce the eigenvalue BVP of Unit 17, and the initial profile is matched by a Fourier series. The **Laplacian** $\nabla^2 u$ governs the three classical equations — heat (diffusion), wave (vibration), and Laplace (steady state).

- **Fourier intuition** — any reasonable periodic signal is a superposition of pure frequencies; the Fourier transform is the continuous-spectrum limit of the Fourier series.
- **Computing the series** — $a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx$, $b_n = \frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx$, derived from trig orthogonality.
- **Symmetry and half-range** — even $\Rightarrow$ cosine series, odd $\Rightarrow$ sine series; extend a function on $0 \leq x \leq L$ evenly or oddly to get either.
- **Separation of variables** — $u(x, t) = X(x)T(t)$ turns a linear homogeneous PDE into two ODEs; the spatial ODE is the Unit 17 eigenvalue problem.
- **The Laplacian and the classical PDEs** — $\nabla^2 u = u_{xx} + u_{yy}$ drives heat $u_t = \alpha^2 u_{xx}$, wave $u_{tt} = c^2 u_{xx}$, and Laplace $\nabla^2 u = 0$.

Part I — Condensed Cheat Sheet

18.1 Fourier Series and Transform Intuition

A periodic function as a superposition of pure-frequency sines and cosines; the time-domain versus frequency-domain picture; harmonics and the fundamental frequency; the Fourier transform as the continuous-spectrum limit of the Fourier series.

The Fourier Series — Superposition of Harmonics

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right), \quad \text{period } 2L.$$

Harmonics and the Fundamental Frequency

the n th harmonic has frequency $\frac{n}{2L}$;
 $n = 1$ is the *fundamental*, $n \geq 2$ the *overtones*.

The Fourier Transform — Continuous Spectrum

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx;$$

the $L \rightarrow \infty$ limit turns the discrete harmonics $n/2L$ into a continuum ξ .

Method — Read a Function in the Frequency Domain

1. Treat the signal $f(x)$ as living in the *time* (or space) domain — its value at each point.
2. Ask which pure frequencies $\cos(n\pi x/L)$, $\sin(n\pi x/L)$ it is built from, and with what amplitudes a_n, b_n .
3. Plot the amplitudes against frequency: this *frequency-domain* spectrum is an equivalent description of the same signal.
4. For a non-periodic signal, let the period $2L \rightarrow \infty$; the discrete spectrum fills in to a continuous one and the sum becomes the Fourier transform integral.

Key Takeaway

A Fourier series says any reasonable periodic function is a **superposition of pure frequencies** — a chord of sines and cosines. The same signal then has two equivalent portraits: the familiar **time-domain** graph and the **frequency-domain** spectrum of amplitudes. Letting the period grow without bound turns the discrete spectrum into a continuous one, and the series becomes the Fourier transform.

Common Pitfall

The constant term is $a_0/2$, **not** a_0 — the factor of $\frac{1}{2}$ is what makes the single Euler–Fourier formula $a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx$ work for $n = 0$ as well. Also, a Fourier *series* needs a **periodic** (or finite-interval) function; a non-periodic signal on the whole line requires the Fourier *transform*, not a series.

18.2 Computing Fourier Series

The Euler–Fourier coefficient formulas on $-L \leq x \leq L$; orthogonality of the trigonometric system as their engine; the constant (DC) term $a_0/2$; convergence at jumps to the midpoint of the one-sided limits (Dirichlet conditions).

The Euler–Fourier Coefficient Formulas

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \quad (n \geq 0), \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \quad (n \geq 1).$$

Orthogonality of the Trigonometric System

$$\int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = L \delta_{nm}, \quad \int_{-L}^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = L \delta_{nm}, \quad \int_{-L}^L \cos \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = 0.$$

Convergence at a Jump (Dirichlet)

At a jump x_0 , the series converges to $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$, the midpoint of the one-sided limits.

Method — Compute a Fourier Series

1. Identify the half-period L so the function repeats with period $2L$, and write the target $f = \frac{a_0}{2} + \sum_n (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L})$.
2. Compute $a_0 = \frac{1}{L} \int_{-L}^L f dx$ — twice the average value of f over one period.
3. Compute a_n and b_n from the Euler–Fourier integrals, using integration by parts where f is polynomial times trig.
4. Exploit symmetry to skip half the work: if f is even all $b_n = 0$; if f is odd all $a_n = 0$ (and $a_0 = 0$).
5. State the series and note its value at any jump is the midpoint $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$.

Key Takeaway

The Euler–Fourier formulas are pure **orthogonality**: integrating f against a single basis wave annihilates every other term and leaves one coefficient. The constant $a_0/2$ is the function’s **average**; the a_n, b_n are how strongly each harmonic is present. Symmetry is the great labor-saver — recognizing f as even or odd erases half the integrals before you compute anything.

Common Pitfall

Watch the normalizations: the coefficients use $\frac{1}{L}$ (over the full period $2L$), while the half-range formulas of the next module use $\frac{2}{L}$ (over $0 \leq x \leq L$). **Mixing the two** is the most common Fourier error. And the series equals f only where f is continuous — at a **jump** it converges to the midpoint $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$, not to either one-sided value.

18.3 Fourier Series for Differential Equations

Even and odd functions and their product rules; the pure cosine series (even) and pure sine series (odd); half-range expansions via even or odd extension of a function given only on $0 \leq x \leq L$;

choosing the expansion that matches the boundary conditions of a PDE.

Even/Odd Symmetry Collapses the Series

f even $\Rightarrow b_n = 0$ (pure cosine series); f odd $\Rightarrow a_n = 0$ (pure sine series, and $a_0 = 0$).

Half-Range Cosine Series (Even Extension)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \quad (n \geq 0).$$

Half-Range Sine Series (Odd Extension)

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

Method — Build a Half-Range Expansion

1. You are given f only on $0 \leq x \leq L$ and must represent it by sines *or* cosines alone.
2. For a *sine* series, extend f as an *odd* function to $-L \leq x \leq 0$; for a *cosine* series, extend it as an *even* function.
3. Apply the half-range formula with the $\frac{2}{L}$ normalization (the $-L$ to 0 contribution doubles the 0 to L integral).
4. Let the boundary conditions of the target PDE pick the series: Dirichlet ends ($u = 0$) call for the *sine* series, Neumann ends ($u_x = 0$) for the *cosine* series.
5. Write the expansion, remembering the odd extension forces $f(0) = f(L) = 0$ in the limit, matching homogeneous Dirichlet data.

Key Takeaway

Symmetry is a choice you can **impose**. A function living only on $0 \leq x \leq L$ has no symmetry of its own, but you may extend it **evenly** to get a cosine series or **oddly** to get a sine series. Which you pick is dictated by the PDE: **Dirichlet** boundary data wants the **sine** series (it vanishes at the ends), **Neumann** data wants the **cosine** series (its slope vanishes).

Common Pitfall

The half-range formulas carry $\frac{2}{L}$, **not** $\frac{1}{L}$ — the odd or even extension doubles the integral over $0 \leq x \leq L$. Do not also integrate over $-L$ to L ; that double-counts. And the sine series of a function with $f(0) \neq 0$ (like $f = 1$ above) **converges to 0 at the endpoints** because the odd extension forces a jump there — expected, not an error.

18.4 Separable Partial Differential Equations

The separation-of-variables ansatz $u(x, t) = X(x)T(t)$; splitting a linear homogeneous PDE into two ODEs joined by a separation constant; the role of homogeneous boundary conditions in producing the spatial eigenvalue BVP; the classical PDEs and their boundary/initial data.

The Separation Ansatz and Constant

$$u(x, t) = X(x)T(t);$$
$$\frac{1}{\alpha^2} \frac{T'}{T} = \frac{X''}{X} = -\lambda \quad (\text{constant: each side depends on a different variable}).$$

The Two Resulting ODEs (Heat Equation)

$$X'' + \lambda X = 0 \quad (\text{spatial, an eigenvalue BVP}); \quad T' + \alpha^2 \lambda T = 0 \quad (\text{temporal, first order, } T_n = e^{-\alpha^2 \lambda_n t}).$$

Superposition of the Product Solutions

$$u(x, t) = \sum_{n=1}^{\infty} b_n X_n(x) T_n(t);$$

the b_n are fixed by matching the initial condition with a Fourier series.

Method — Solve a PDE by Separation of Variables

1. Assume a product solution $u(x, t) = X(x)T(t)$ and substitute into the PDE.
2. Divide through so all x -dependence is on one side and all t -dependence on the other; set both equal to the separation constant $-\lambda$.
3. Solve the *spatial* ODE with the homogeneous boundary conditions first — this is the Unit 17 eigenvalue BVP, yielding eigenvalues λ_n and eigenfunctions X_n .
4. Solve the *temporal* ODE for each λ_n to get $T_n(t)$, then form the superposition $u = \sum_n b_n X_n T_n$.
5. Impose the initial condition $u(x, 0) = f(x)$ and read off the b_n as the Fourier (half-range) coefficients of f .

Key Takeaway

Separation of variables is the master move for linear PDEs: assuming $u = X(x)T(t)$ **decouples** the equation into two ODEs tied by a single separation constant. The homogeneous boundary conditions turn the spatial ODE into the **eigenvalue BVP of Unit 17**, so its eigenfunctions become the building blocks; **superposition** assembles them and the **initial condition** fixes the weights through a Fourier series.

Common Pitfall

Separation of variables needs the PDE to be **linear and homogeneous with homogeneous boundary conditions** — a nonzero boundary value or a forcing term breaks the clean split (handle those by first subtracting a steady-state/particular solution). Also, always solve the **spatial** eigenvalue problem before the temporal one: the boundary conditions live in x , and they are what quantize λ .

18.5 The Laplacian

The Laplacian $\nabla^2 u = \nabla \cdot \nabla u = u_{xx} + u_{yy}$ as the divergence of the gradient; its meaning as how a value departs from its local average; Laplace's equation $\nabla^2 u = 0$ and harmonic functions; the steady-state limit of diffusion; polar/spherical forms.

The Laplacian — Divergence of the Gradient

$$\nabla^2 u = \nabla \cdot (\nabla u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \quad (\text{Cartesian, here through 3D}).$$

Laplace's Equation and Harmonic Functions

$$\nabla^2 u = 0 \quad \Longleftrightarrow \quad u \text{ is harmonic;}$$

value at a point = average over any surrounding circle/sphere (mean value).

The Laplacian in Polar Coordinates

$$\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} \quad (\text{the natural form for disks and annuli}).$$

Method — Test and Interpret the Laplacian

1. Compute the second partial in each spatial variable and add them to form $\nabla^2 u$.
2. Read the sign as curvature versus the neighborhood: $\nabla^2 u > 0$ means the value sits *below* its local average (a dip), $\nabla^2 u < 0$ means *above* (a bump).
3. If $\nabla^2 u = 0$ everywhere, the function is harmonic — it equals its local average and has no interior maxima or minima (maximum principle).
4. For diffusion, recognize $\nabla^2 u$ as the spatial driver: heat flows so as to erase departures from the average, and the steady state ($u_t = 0$) is exactly Laplace's equation.

Key Takeaway

The Laplacian $\nabla^2 u$ is the **divergence of the gradient** — a scalar measuring how a point's value departs from the **average of its neighbors**. Where it is positive the point is a relative dip, where negative a relative bump, and where it is **zero the function is harmonic** and equals its local average. This single operator is the spatial engine of all three classical PDEs.

Common Pitfall

The Laplacian is a *sum* of unmixed second partials, $u_{xx} + u_{yy}$ — **do not include the mixed partial** u_{xy} , and do not confuse $\nabla^2 u$ (a scalar) with the gradient ∇u (a vector) or the Hessian (a matrix). In curved coordinates you **cannot** just sum naive second derivatives: polar form carries the extra $\frac{1}{r}u_r$ and $\frac{1}{r^2}u_{\theta\theta}$ terms.

18.6 The Heat and Wave Equations

The heat equation $u_t = \alpha^2 u_{xx}$ (parabolic diffusion) solved by separation plus a Fourier sine series; the wave equation $u_{tt} = c^2 u_{xx}$ (hyperbolic vibration) with standing-mode and d'Alembert traveling-wave solutions; Duhamel's principle for a forced wave; the classification of the three classical equations.

The Three Classical Equations

Heat (parabolic): $u_t = \alpha^2 u_{xx}$;

Wave (hyperbolic): $u_{tt} = c^2 u_{xx}$;

Laplace (elliptic): $u_{xx} + u_{yy} = 0$.

Heat Equation — Fourier Solution

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

Wave Equation — d'Alembert's Solution

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds \quad (\text{initial shape } f, \text{ initial velocity } g).$$

Method — Solve the 1D Heat Equation on a Rod

1. Separate $u = X(x)T(t)$ and solve the spatial BVP $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ to get $\lambda_n = (n\pi/L)^2$, $X_n = \sin \frac{n\pi x}{L}$.
2. Solve $T' + \alpha^2 \lambda_n T = 0$ for $T_n = e^{-\alpha^2 (n\pi/L)^2 t}$, so each mode decays faster the higher its frequency.
3. Superpose: $u(x, t) = \sum_n b_n \sin \frac{n\pi x}{L} e^{-\alpha^2 (n\pi/L)^2 t}$.
4. Set $t = 0$: $u(x, 0) = \sum_n b_n \sin \frac{n\pi x}{L} = f(x)$, so the b_n are the half-range *sine* coefficients of the initial temperature f .
5. Evaluate $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$ and write the full series solution.

Key Takeaway

The classical PDEs all yield to the same pipeline: **separate**, solve the **spatial eigenvalue BVP**, attach the time factor, **superpose**, and fit the initial data with a **Fourier series**. Their characters differ — the **heat** equation smooths and decays (high modes die fastest), the **wave** equation oscillates and propagates (d'Alembert's two traveling waves), and **Laplace** gives the steady state — but Fourier analysis is the common engine.

Common Pitfall

Match the time factor to the equation's order in t : the heat equation is *first* order so each mode carries a single decaying $e^{-\alpha^2 \lambda_n t}$, while the wave equation is *second* order so each mode needs **two** time functions $\cos(c\sqrt{\lambda_n} t)$ and $\sin(c\sqrt{\lambda_n} t)$ — and therefore *two* initial conditions (shape *and* velocity). Using a single exponential for the wave equation **loses half the solution**.

Unit 18 Cheat-Sheet Takeaway

Fourier analysis is the through-line. A periodic function is a sum of harmonics with coefficients a_n, b_n pulled out by **orthogonality**; **even/odd symmetry** reduces it to a pure cosine or sine series, and half-range extensions let you choose. That same series solves PDEs: **separation of variables** $u = X(x)T(t)$ splits the equation into ODEs, the spatial part is the Unit 17 **eigenvalue BVP**, and the initial data is matched by a Fourier series. The **Laplacian** $\nabla^2 u$ powers the heat ($u_t = \alpha^2 u_{xx}$), wave ($u_{tt} = c^2 u_{xx}$), and Laplace ($\nabla^2 u = 0$) equations.